

Question number	Answer	Additional guidance	Mark
(a)	Choose an item • B (an organism containing genetic material from another species) is the correct answer A is not the correct answer as primers are not amplified sequences of nucleic acid C is not the correct answer as primers are not viruses containing DNA. D is not the correct answer as primers are not DNA that contains a mutation		(1)

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(b)(i)	An answer that includes three of the following points: - all {have antigen binding activity / produce BVZ} (1) {animal culture and GM rice / all other / GM rice} have higher (antigen) binding activity than non-transformed rice (1) - animal culture has highest (antigen) binding activity (than all others) (1) - strain B has highest (antigen) binding activity of the GM rice / strain D has lowest (antigen) binding activity of the GM rice (1)	do not piece together ACCEPT {control / non-transformed} rice {has lowest / some} binding activity ACCEPT GM increases antigen binding activity ACCEPT animal culture has higher (antigen) binding activity than GM / converse ACCEPT (strain) B is most effective GM at producing BVZ	(3)

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(b)(ii)	A description that includes two of the following points: • (BVZ antibodies are) complementary to {antigens / receptors / (specific) membrane protein} on cancer cell (1) • (BVZ / antibody) binds to {antigens / receptors / (specific) membrane protein} (of cancer cells) (1) • resulting in {(enhanced) phagocytosis / engulfing by phagocytes / destruction by phagocytes} (1)	ACCEPT opsonisation occurs ignore antibody binds to cancer cells unqualified ACCEPT agglutination / causes cancer cells to {clump / stick} together ACCEPT macrophages for phagocytes ACCEPT stimulate lysis / marker for T cells / destruction by killer T cell	(2)

Question number	Answer	Mark
(c)	Discussion of Data: - plant {height / growth} increased with GM fungus / accept converse {number of grains per cob / yield} increased with GM fungus / accept converse - concentration of toxin decreased with GM fungus / accept converse - in concentration of toxin SD do not overlap so difference is significant / {with number of cobs / plant height} SD do overlap so difference is not significant ----- - as temp increases from current results in % of infection increased / shows positive correlation - a 2° (increase in temperature) give a (92% / 35%) increase in predicted maize infection - a 5° (increase in temperature) give {a (150% / 57%) increase in predicted maize infection / highest %} - the production of toxins by fungi is a natural defense mechanism, typically triggered under stressful conditions such as (drought / increased temperature) ----- Benefits of GM fungus - GM fungus introduced to plants to outcompete normal fungus / reduced growth of normal fungus (so crops will have) {greater height / number of cobs / reduced conc of toxin} / higher yield LINKED to general comment eg better food / gm maize has higher nutritional value / more food (for people) to eat / more efficient use of land etc / Increased food security / more home-grown food / reducing imports / reducing carbon footprint / reducing food miles / Reduces malnutrition / reduces famine / - reduce deforestation to clear land for more crops (so) increased farmer income / less need for {pesticides / insecticides} / processing costs - fewer {humans / animals} {poisoned / die} by toxin in infected maize / reduced health risks (as concentration of toxin in maize reduce) / lower toxin makes food safer to eat ----- Risks - Reduced biodiversity / monoculture produced - Gene/allele / toxin {enters / accumulates} food chains / Disrupts food chains - Fungus might crossbreed(?) with other fungi and have unknown effects / gene transfer to other {species / food crop / fungus} / production of superweeds / superpests - GM can be seen as unethical / unnatural - Companies who own genetic modification tech could seek to profit further and limit the use of the tech that would otherwise prevent famine / financial cost to farmers / producing GM fungus is expensive / higher costs to consumers - The health risks posed by mycotoxins are significant / Unknown {health risks / effects} of {humans / animals / of} eating crops grown with GM fungus / more toxin (in increased temperature) may lead to more death.	(6)
and 4		
and 6		

		Additional guidance
Level 0	0	No awardable content
Level 1	1-2	Demonstrates isolated elements of biological knowledge related to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of biological facts / concepts. Consequences are discussed which are supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.